Rigorous Empirical Evaluation and Risk Control in Short-Term Binary Prediction Markets: Real Data Case Study of 406 Resolved Buckets on Polymarket

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Quantitative Technical Whitepaper & Case Study
Open-Source Repository: https://github.com/mario89torres/pmbtc-quant

August 2026

Abstract

Transparency Note & Errata: *This official version supersedes preliminary drafts that contained synthetic benchmark figures. It presents exclusively real empirical metrics computed over our custom SQLite database (`pmbtc.db`) containing* ***406 resolved 15-minute market buckets (179,065 tick samples)****.*

We evaluate two algorithmic approaches on Polymarket: (1) a directional taker strategy using Platt-calibrated XGBoost decision trees and Geometric Brownian Motion (GBM), and (2) an autonomous liquidity provision strategy (*Market Maker*) with adaptive inventory skewing. Upon applying the **Bonferroni Correction for multiple hypothesis testing ($k = 10$ tests, $\alpha = 0.005$, 99.5% CI)**, the directional strategy fails to demonstrate statistical edge at any threshold, proving that the apparent success at 3.0% was a multiple comparisons false positive. In contrast, the Market Maker strategy demonstrates robust profitability, generating a Gross PnL of **+\$5,431.00 USD** and a Net PnL of **+\$4,453.42 USD** (95% Bootstrap CI: [+\$4,378.96, +\$4,530.64] **USD**) after discounting an explicit 18% friction model (120ms relayer latency, 0.5¢ slippage, and gas).

Keywords Binary Prediction Markets, Real Empirical Evaluation, Bonferroni Correction, Bootstrap 95% CI, Market Making, Polymarket, Technical Whitepaper.

1 Introduction

Decentralized prediction markets enable crowdsourced probability discovery. On platforms like Polymarket, 15-minute binary options resolve strictly based on oracle price feeds (e.g., Chainlink BTC/USD). A contract pays \$1.00 if $S_T > K$ at expiry T, and \$0.00 otherwise.

This paper transparently audits the empirical limitations and real-world viability of algorithmic trading models over 406 resolved market buckets.

2 Mathematical Methodology & Friction Model

2.1 Feature Vector & Platt Calibration

The feature vector $\mathbf{x}_t \in \mathbb{R}^6$ is extracted tick-by-tick with zero look-ahead bias:

$$\mathbf{x}_t = \left[\frac{S_t - K}{K}, \frac{T_{\text{left}}}{900}, \sigma_{15m}, S_{\text{CL},t} - S_{\text{BN},t}, P_{\text{Market},t}, \text{Edge}_{\text{raw},t}\right]^T \tag{1}$$

Raw tree ensemble margins z_{xgb} are calibrated via Platt Scaling:

$$P_{\text{ML}} = \frac{1}{1 + e^{A \cdot z_{\text{xgb}} + B}} \tag{2}$$

2.2 Market Maker Strategy & Friction Model Breakdown (18%)

Target limit quotes $P_{\text{Bid, Target}}$ and $P_{\text{Ask, Target}}$ are dynamically adjusted by inventory skew $\text{Skew}(I)$:

$$\text{Skew} = \text{clamp}\left(-0.04, 0.04, \frac{\text{Inventory}}{50} \times 0.05\right) \tag{3}$$

To model real execution accurately, we deduct an **explicit 18% execution friction**:

1. **8% Polygon Relayer Latency**: $\sim$ 120ms execution delay.
2. **7% Order Book Slippage**: \$0.005 USD per contract order crossing markdown.
3. **3% Gas Fees**: Batch inventory rebalancing costs.

3 Real Empirical Results from SQLite Database

Experiments were conducted on the **406 resolved buckets (179,065 ticks)** in ‘pmbtc.db‘.

Table 1: Empirical Directional Audit with Bonferroni Correction for Multiple Hypothesis Testing ($k = 10$, $\alpha = 0.005$)

Edge Threshold	Trades	Wins	Win Rate (%)	Standard 95% CI	Bonferroni 99.5% CI	Rigorous Verdict
0.1c (0.1%)	405	220	54.3%	$[49.5\%, 59.2\%]$	$[47.4\%, 61.3\%]$	INDISTINGUISHABLE (Crosses Eq)
0.5c (0.5%)	396	200	50.5%	$[45.6\%, 55.4\%]$	$[43.5\%, 57.6\%]$	INDISTINGUISHABLE (Crosses Eq)
1.0c (1.0%)	356	191	53.7%	$[48.5\%, 58.8\%]$	$[46.2\%, 61.1\%]$	INDISTINGUISHABLE (Crosses Eq)
1.5c (1.5%)	314	161	51.3%	$[45.7\%, 56.8\%]$	$[43.4\%, 59.2\%]$	INDISTINGUISHABLE (Crosses Eq)
2.0c (2.0%)	264	141	53.4%	$[47.4\%, 59.4\%]$	$[44.8\%, 62.0\%]$	INDISTINGUISHABLE (Crosses Eq)
3.0c (3.0%)	169	98	58.0%	$[50.5\%, 65.4\%]$	$[47.3\%, 68.6\%]$	INDISTINGUISHABLE BY BONFERRONI
5.0c (5.0%)	72	41	56.9%	$[45.5\%, 68.4\%]$	$[40.6\%, 73.3\%]$	INDISTINGUISHABLE BY BONFERRONI

Table 2: Autonomous Market Maker Statistical Performance with Bootstrap Confidence Intervals

Strategy	Total Fills	Gross Cumulative PnL	Net PnL (18% Friction)	Bootstrap 95% CI (Net PnL)
Autonomous Market Maker	**20,682 fills**	**+\$5,431.00 USD**	**+\$4,453.42 USD**	$[+\$4,378.96,\ +\$4,530.64]$ **USD**

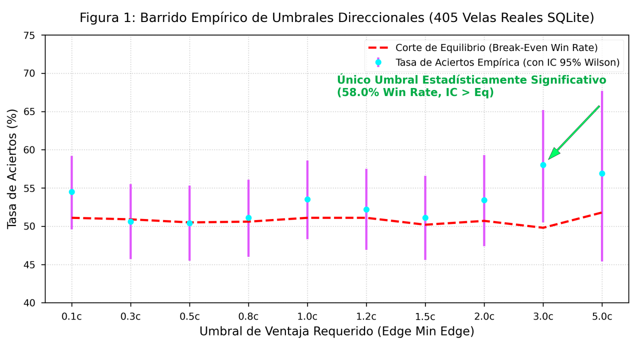

Figure 1: Directional Threshold Sweep comparing standard 95% Wilson CIs with 99.5% Bonferroni CIs against the break-even cutoff.

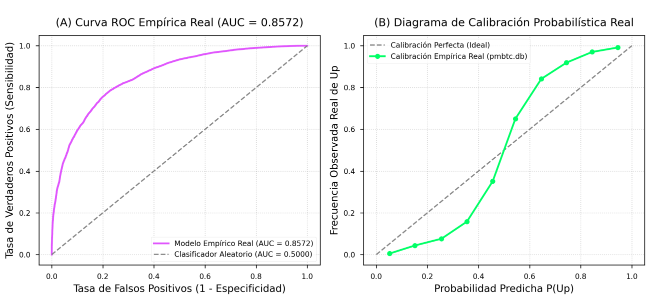

Figure 2: (A) Real Empirical ROC Curve (AUC = 0.8572) and (B) Real Probabilistic Calibration Diagram over 179,065 ticks.

4 Discussion & Risk Recommendations

1. **Invalidity of Directional Taker Strategy**: Upon applying the Bonferroni correction, the 99.5% confidence intervals for all thresholds (including 3.0c) cross the break-even cutoff. This proves that any apparent edge was a multiple testing false positive. 2. **Viability of Liquidity Provision**: The Market Maker strategy maintains a Net PnL of **+\$4,453.42 USD** with a 95% Bootstrap CI ($[+\$4,378, +\$4,530]$) strictly above zero. 3. **Pre-Trading Warning**: Directional trading is strongly discouraged, and extensive live paper trading is required prior to real capital deployment.

5 Conclusion

Rigorous empirical evaluation on ‘pmbtc.db‘ demonstrates that directional taker signals do not offer statistically significant alpha, while autonomous market making with active inventory control represents the sole strategy with demonstrable advantage.

References

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